

TW06: pandas and DataFrames

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Create, inspect, select, clean, transform, group, and merge labelled tabular data with pandas.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Create and inspect labelled data

Scenario

A sales file must be understood before any business calculation begins.

INSTRUCTIONS

1. Read `beginner_sales.csv` into a `DataFrame`.
2. Display shape, column names, dtypes, first rows, and last rows.
3. Create a `Series` containing `unit_price` with `order_id` as its index.
4. Use `describe()` for numerical columns and include category counts separately.
5. Write two observations about structure or quality without changing the data.

REQUIRED SUBMISSION

- A loaded `DataFrame` and indexed `Series`.
- Complete structural and descriptive inspection.
- Two accurate observations.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Select and filter rows safely

Scenario

A manager wants a focused view of selected columns, positions, and business conditions.

INSTRUCTIONS

1. Load the sales data and set `order_id` as the index while keeping a raw copy.
2. Use `.loc` to select two named orders and chosen columns.
3. Use `.iloc` to select the first three rows and first four columns.
4. Filter orders from the North with `unit_price` above 20.
5. Create a revenue column on a copy and verify the raw table is unchanged.

REQUIRED SUBMISSION

- One `.loc` result, one `.iloc` result, and one filtered result.
- A transparent revenue calculation.
- A passing raw-data preservation check.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Clean missing, duplicated, and inconsistent data

Scenario

The sales file contains blank values, duplicate rows, and region labels with inconsistent spacing and capitalisation.

INSTRUCTIONS

1. Profile missing values, exact duplicates, and unique region labels.
2. Remove exact duplicates while recording how many were removed.
3. Standardise region and product text with string methods.
4. Convert units and satisfaction to numeric with invalid values coerced to missing.
5. Choose and justify a treatment for each missing field, then validate the result.

REQUIRED SUBMISSION

- A before-and-after quality table.
- A cleaned DataFrame with documented decisions.
- At least four passing assertions.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Parse dates, group results, and merge tables

Scenario

Sales must be summarised by month and linked with customer information.

INSTRUCTIONS

1. Parse order_date and list any failed conversions.
2. Create month and weekday columns.
3. Calculate revenue and group total revenue by region and month.
4. Load customers.csv and merge it onto sales using customer_id.
5. Use a merge indicator to audit matched and unmatched customer keys.

REQUIRED SUBMISSION

- A date-failure audit and two calendar features.
- A region-by-month summary.
- A left merge with match-quality counts.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Deliver a beginner sales report

Scenario

A business owner needs one reproducible table answering what sold, where, and with what customer satisfaction.

INSTRUCTIONS

1. Load and clean only the fields needed for the report.
2. Calculate valid-order count, total units, total revenue, average order revenue, and median satisfaction.
3. Create region and product summaries with groupby and agg.
4. Sort each summary by the most relevant measure.
5. Write a finding, limitation, and next question supported by the output.

REQUIRED SUBMISSION

- Five labelled key performance indicators.
- Two sorted grouped summary tables.
- An evidence-based finding, limitation, and next question.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____